

PUZZLES OF SURVIVAL

split brains

my division



We now have three “pure” views of personal identity on the table, along with the various combinations of those views.

Our topic today is a cluster of puzzles about identity and survival. These puzzles put pressure, in different ways, on some of the assumptions about survival we have been making so far.

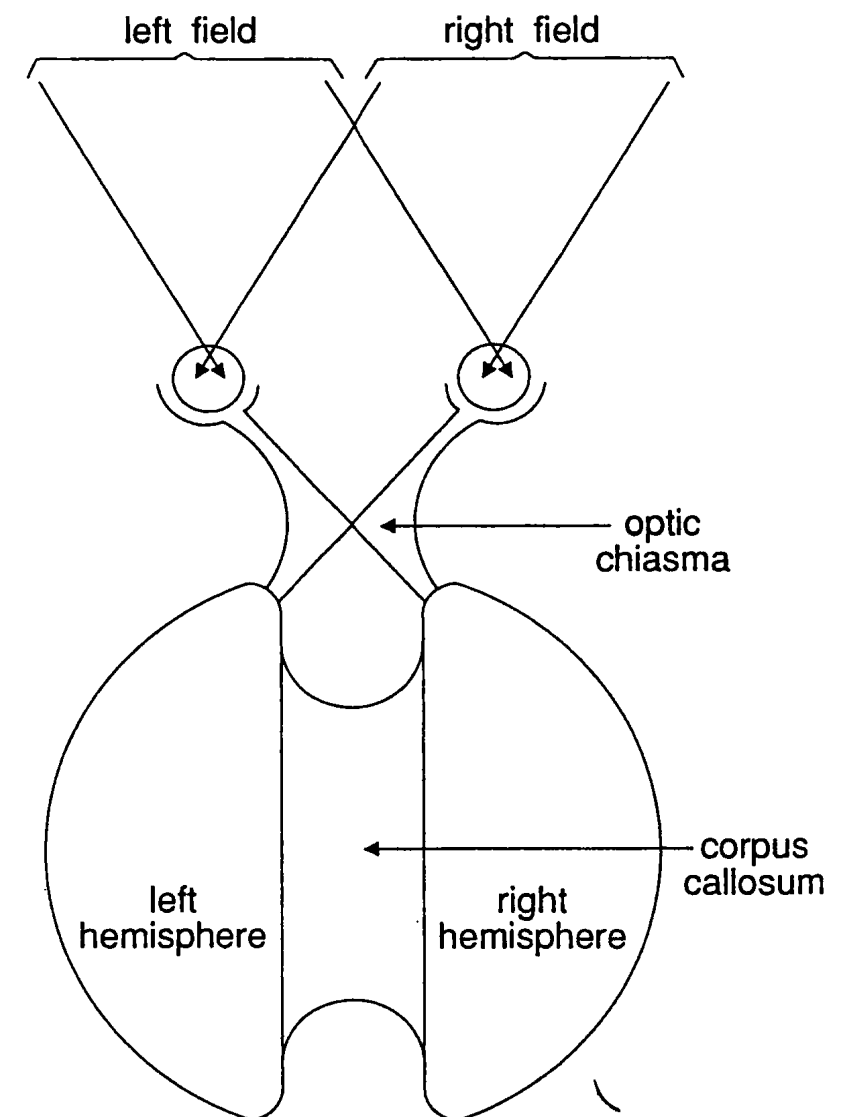
One good way to test out your preferred view of survival is to think about what your view implies with respect to these puzzles.

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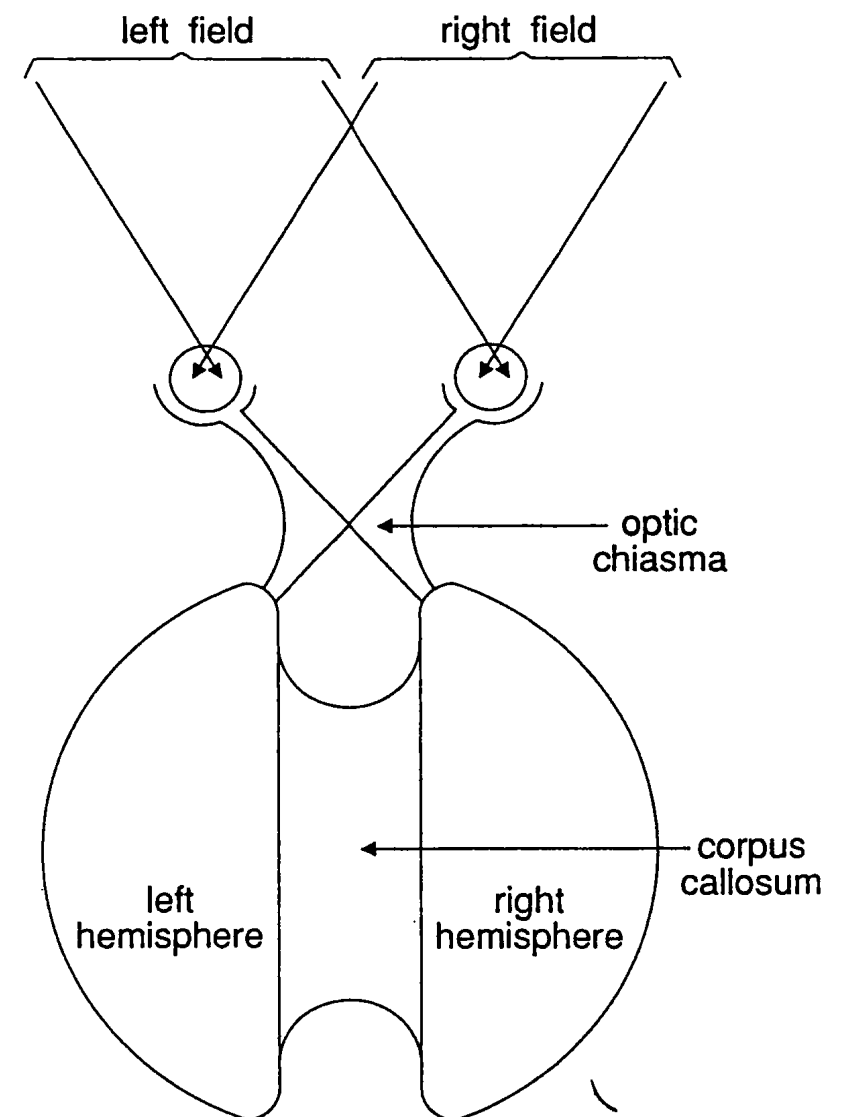
These are patients whose corpus callosum has been severed. The corpus callosum is a pathway which connects the left and right hemispheres of the human brain and, in normal subjects, allows the two hemispheres of the brain to exchange information.

If the corpus callosum is severed, the two hemispheres of the brain cannot exchange information. So any sensory data about the environment available to, for example, the left hemisphere, will not be available to guide the movements of the left hand, which is controlled by the right hemisphere. Information available only to the right hemisphere will not be reportable in speech, since speech is controlled by the left hemisphere.



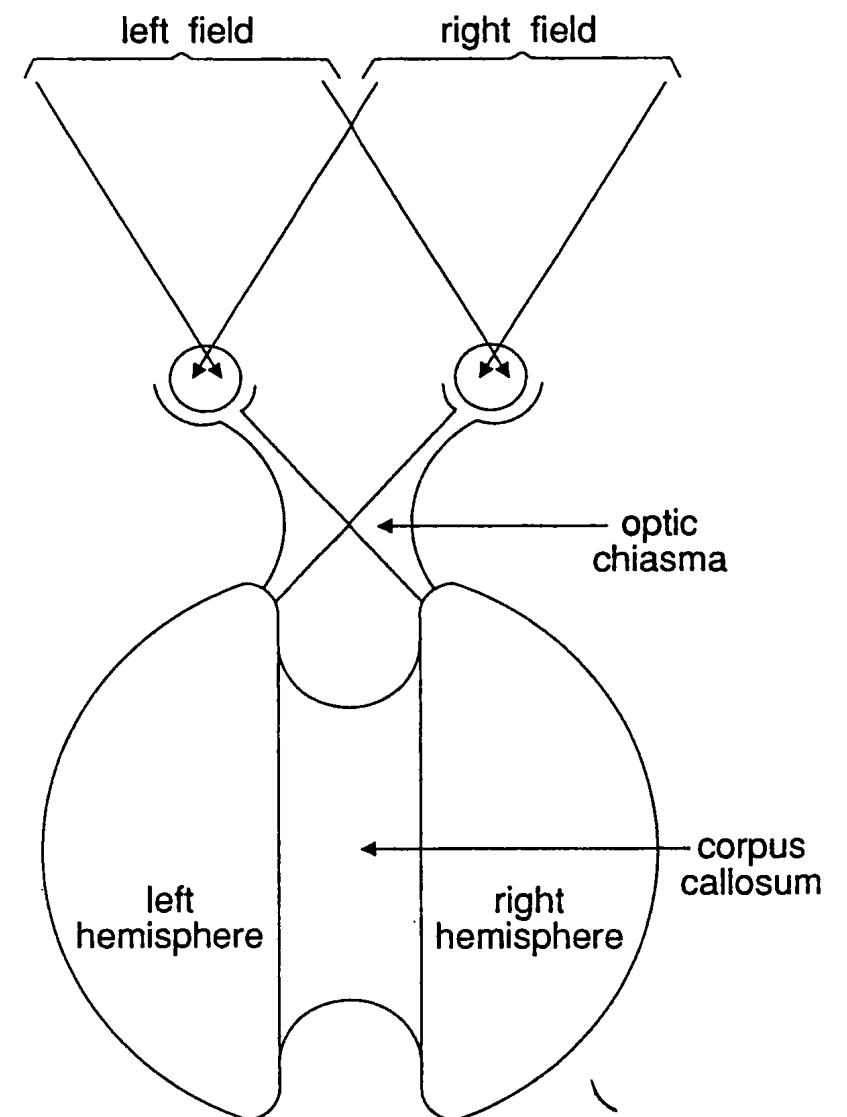
The results of giving sensory data to just one of the hemispheres of the brain of such a patient are striking.

“What is flashed to the right half of the visual field, or felt unseen by the right hand, can be reported verbally. What is flashed to the left half of the visual field ... cannot be reported, though if the word ‘hat’ is flashed on the left, the left hand will retrieve a hat from a group of concealed objects if the person is told to pick out what he has seen. At the same time, he will insist verbally that he saw nothing.”



The results of giving sensory data to just one of the hemispheres of the brain of such a patient are striking.

“... if two different words are flashed to the two half fields (e.g., ‘pencil’ and ‘toothbrush’) and the individual is told to retrieve the corresponding object from beneath a screen, with both hands, then the hands will search the collection of objects independently, the right hand picking up the pencil and discarding it while the left hand searches for it, and the left hand similarly rejecting the toothbrush which the right hand lights upon with satisfaction.”



Why are these cases puzzling from the point of view of theories of survival?

Think about a case in which a split-brain patient has a red stimulus presented to the right half of their visual field, and a blue stimulus presented to the left half of their visual field. If you ask the subject what color they see, they will say “Red”, since this was the color presented to the part of the eye which feeds input to the left hemisphere of the brain, which controls speech. If you ask the person to pick out with their left hand an object of the same color as the one they saw, they will pick out a blue object.

The following seems clearly to be true:

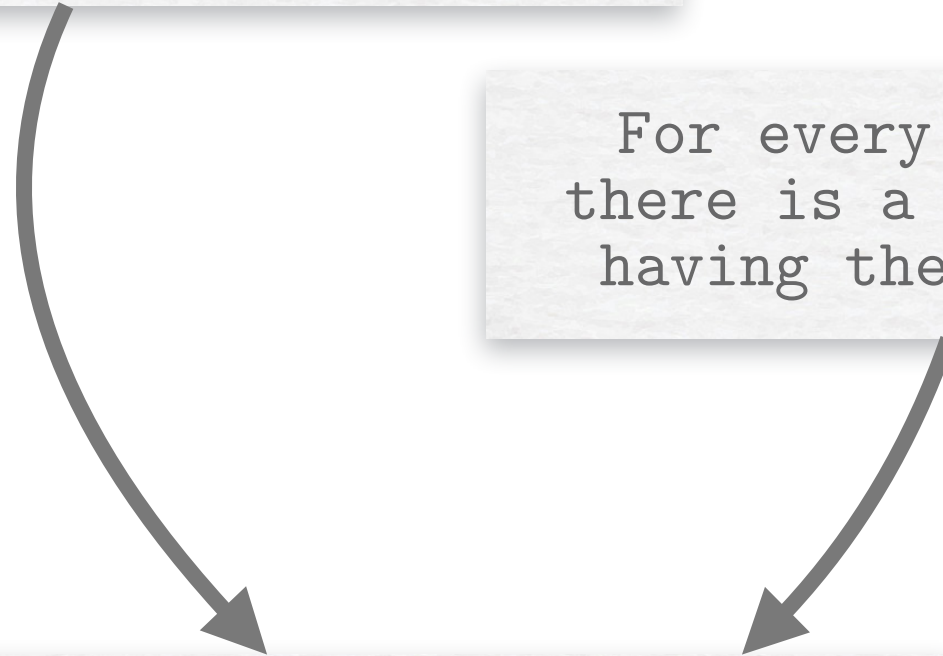
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But the following also seems
clearly to be true:

For every experience,
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But it also seems plausible that if someone is having a conscious experience of a color, they can be aware of that experience.

There is a person having a reddish experience and a person having a bluish experience.

If a person has a conscious reddish (or bluish) experience, it must be possible for them to be aware of that experience.

It is not possible for the person having the reddish experience to be aware of the bluish experience (or vice versa).

In this example, the individual having the reddish experience $\neq$ the person having the bluish experience.

But this implies that, at least while they are being given stimuli of this kind, the bodies of split brain patients are inhabited by two people.

But if this is right, one of the following must be true:

(1) While the split brain patients are in experiments of this sort, there are two persons inhabiting their body; but, at other times, there is just one person inhabiting their body.

(2) Split brain patients always have two persons inhabiting their body, but non-split brain subjects do not.

(3) All of us, split-brain and non-split-brain subjects alike, have two (or more) persons inhabiting their body.

(1) While the split brain patients are in experiments of this sort, there are two persons inhabiting their body; but, at other times, there is just one person inhabiting their body.

If (1) were true, then simply flashing some red and blue lights at someone would bring a new person into existence; and turning off the lights would kill that person.

(2) Split brain patients always have two persons inhabiting their body, but non-split brain subjects do not.

If (2) were true, then severing the corpus callosum of an epileptic patient would bring a new person into existence; and reversing the surgery would kill that person.

(3) All of us, split-brain and non-split-brain subjects alike, have two (or more) persons inhabiting their body.

Non-split brain patients never have conscious experiences of which they are not aware; but then if (3) were true it would follow that there is a person inhabiting my body which never has any conscious experiences at all. But then in what sense does that person even exist?

One kind of response to these cases is to call into question an assumption that we found plausible when introducing the survival question: the assumption that there must always be a determinate fact of the matter about whether X and Y are the same person.

It is at least somewhat plausible that the corresponding assumption is false about some complex entities. Consider, for example, sports teams. We might ask whether the Notre Dame football team in week 1 of next year's season is **really the same team as** the team in week 5.

In one sense, they are clearly the same team. But we can also suppose that they have had some injuries and transfers, so that the players are not the same. In that sense, they are different teams. Here there is some temptation to think that the question, 'Is this numerically one team?' has no determinate answer.

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Some people find the split brain cases puzzling enough that they are tempted to say the same sort of thing about persons. On this sort of view, the question 'Is the person having the reddish experience numerically the same person as the one having the bluish experience?' might have no determinate answer. In one sense there is one person, there; in another sense, there are two people there.

Others find this view about survival deeply counterintuitive.

Split brain cases are puzzling no matter what your answer to the survival question is.

Our second puzzle targets more directly certain answers to the survival question.

Let's return to a case we discussed earlier: Derek Parfit's example of the Brain Transplant.

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“Suppose first that I am one of a pair of identical twins, and that both my body and my twin's brain have been fatally injured. Because of advances in neuro-surgery, it is not inevitable that these injuries will cause us both to die. We have between us one healthy brain and one healthy body. Surgeons can put these together.

If all of my brain continues both to exist and to be the brain of one living person, who is psychologically continuous with me, I continue to exist. This is true whatever happens to the rest of my body. ...”

As Parfit points out, we can also imagine a case in which only one hemisphere of your brain survives.

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“It is in fact true that one hemisphere is enough. There are many people who have survived, when a stroke or injury puts out of action one of their hemispheres. With his remaining hemisphere, such a person may need to re-learn certain things, such as adult speech, or how to control both hands. But this is possible. ... [So] I would survive if my brain was successfully transplanted into my twin's body. And I could survive with only half my brain, the other half having been destroyed. Given these two facts, it seems clear that I would survive if half my brain was successfully transplanted into my twin's body, and the other half was destroyed.”

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Could you survive Hemisphere Transplant? As Parfit says, we can imagine that there would be more or less complete psychological continuity before and after the surgery. So on the psychological theory of survival, it seems clear that you would survive.

What should the proponent of soul survival say? Unlike the psychological theorist, the soul theorist is not forced into an answer here. But the fact of psychological continuity would seem to make the view that one can survive attractive.

What should the proponent of a materialist theory say? We saw that the case of Brain Transplant was reason for the materialist to say that survival requires the survival of a brain rather than a whole organism. Just so, perhaps the case of Hemisphere Transplant suggests that survival only requires the survival of one hemisphere of the brain.

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This leads to a case about which it is really hard to know what we should say.

“My Division. My body is fatally injured, as are the brains of my two brothers. My brain is divided, and each half is successfully transplanted into the body of one of my brothers. Each of the resulting people believes that he is me, seems to remember living my life, has my character, and is in every other way psychologically continuous with me. And he has a body that is very like mine.”

Let's call the resulting people Lefty and Righty. Then we seem to have four options.

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You survive
as Lefty.

You survive
as Righty.

You survive as
Lefty and Righty.

You do not
survive.

It is hard to see how the first two could be true, since nothing seems to favor one over the other.

But the third cannot be true, since Lefty $\neq$ Righty.

So it looks like the last option must be true: you do not survive.

Here's the puzzle: in both Hemisphere Transplant and My Survival, one of your hemispheres is successfully transplanted into a body which goes on living. Given that, how could you survive in one case but not the other?

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This looks puzzling from the point of view of both materialist views and the psychological view. (We'll ask what the believer in souls should say about this case in a second.)

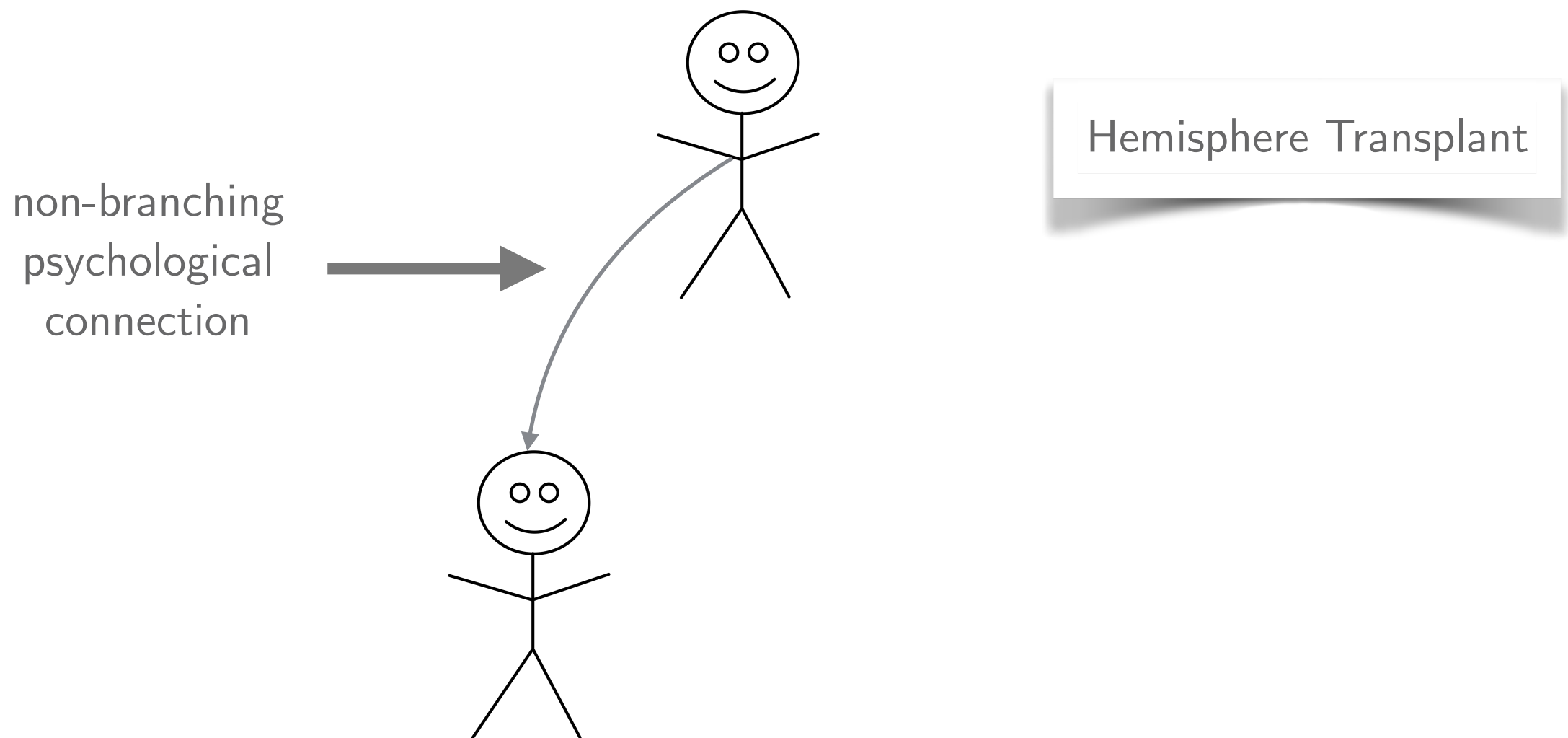
One possibility, which we have already discussed, is to deny that one can survive Hemisphere Transplant. We've already seen some problems with that. A second strategy is to try to find some relevant **difference** between Hemisphere Transplant and My Division which could explain why you survive one but not the other.

What could the relevant difference be? One difference is that in the case of Hemisphere Transplant (but not My Division) the psychological connection between the individuals is a **non-branching connection**. So some psychological theorists say that for you to survive at some later time, you must stand in a certain psychological connection to that thing and **to nothing else**.

This would explain why you can survive Hemisphere Transplant but not My Division.

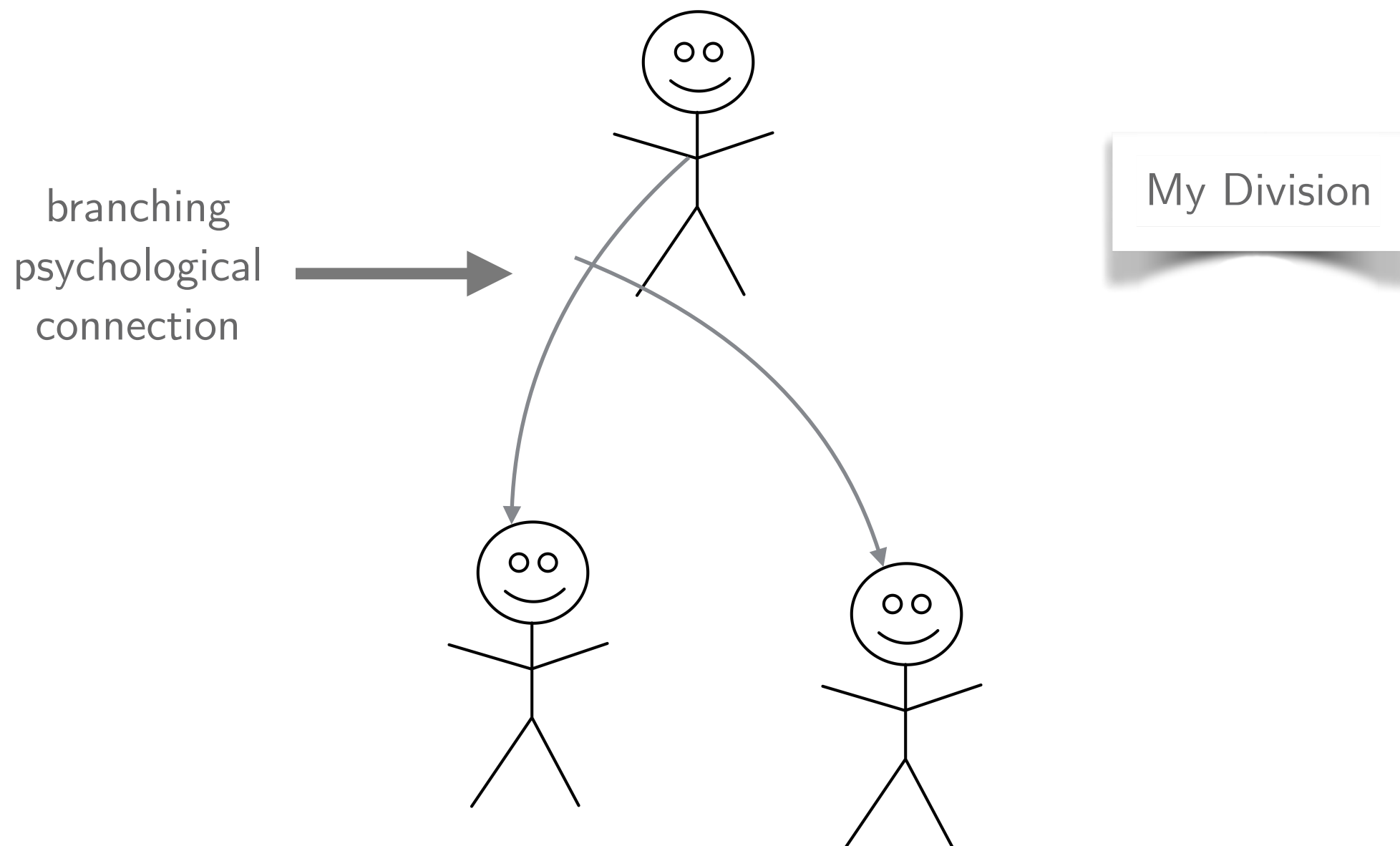
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Notice that a materialist could also adopt this kind of view. They could say that what is required for your survival is a **non-branching material relation** to some future thing.

But here is an odd consequence of this view. Suppose that you wake up, and are told that you are the result of a hemisphere transplant operation. Next to you on another hospital bed is a body into whom the other hemisphere was transplanted.

You do not yet know whether that body will awake.

You think that you know that you are the person you remember being. But, if that other body survives to become a living person, you will not be the person you remember being.

But how could your identity — who you are — depend on what happens with that other body?

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This is an argument against the idea that we can save the psychological theory of survival by adding on a non-branching clause. Intuitively, the idea is that whether you are the same person as X depends **only** on relations between you and X — it can't possibly depend on relations to some other thing!

We can turn this into an argument against the psychological theory:

THE NO-NON-BRANCHING ARGUMENT

- (1) If some version of the psychological theory of survival is true, it must be a “non-branching” theory.
 - (2) Whether I am the same person as X depends only on relations between me and X.
 - (3) No “non-branching” theory can be true. (2)
-
- (C) The psychological theory of survival is false.

We could construct a parallel argument against pure materialist theories or M+P.

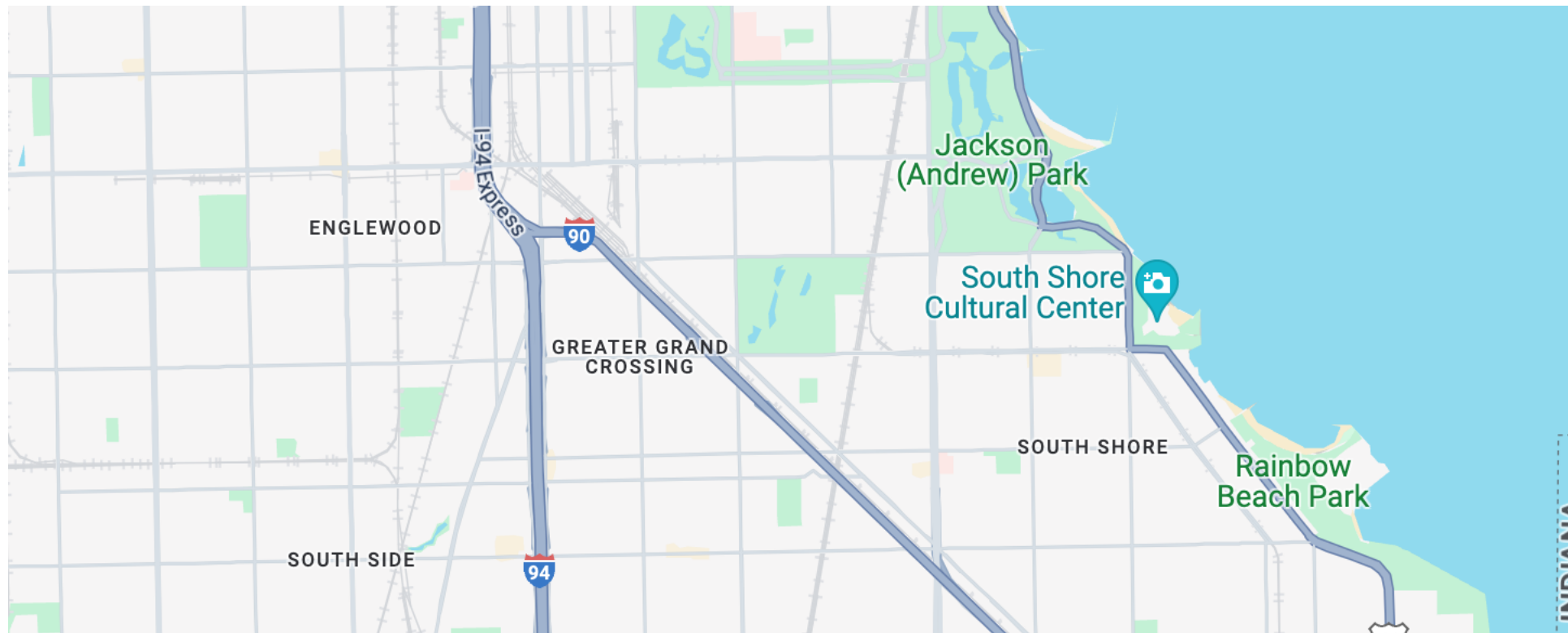
Here's a quite different way of responding to My Division. Recall this option:

You survive as
Lefty and Righty.

We rejected this on the grounds that Lefty is not identical to Righty, so that you cannot be identical to both. But this neglects a possibility: maybe you were already Lefty *and* Righty. You thought that you were one person; but in fact you were already two.

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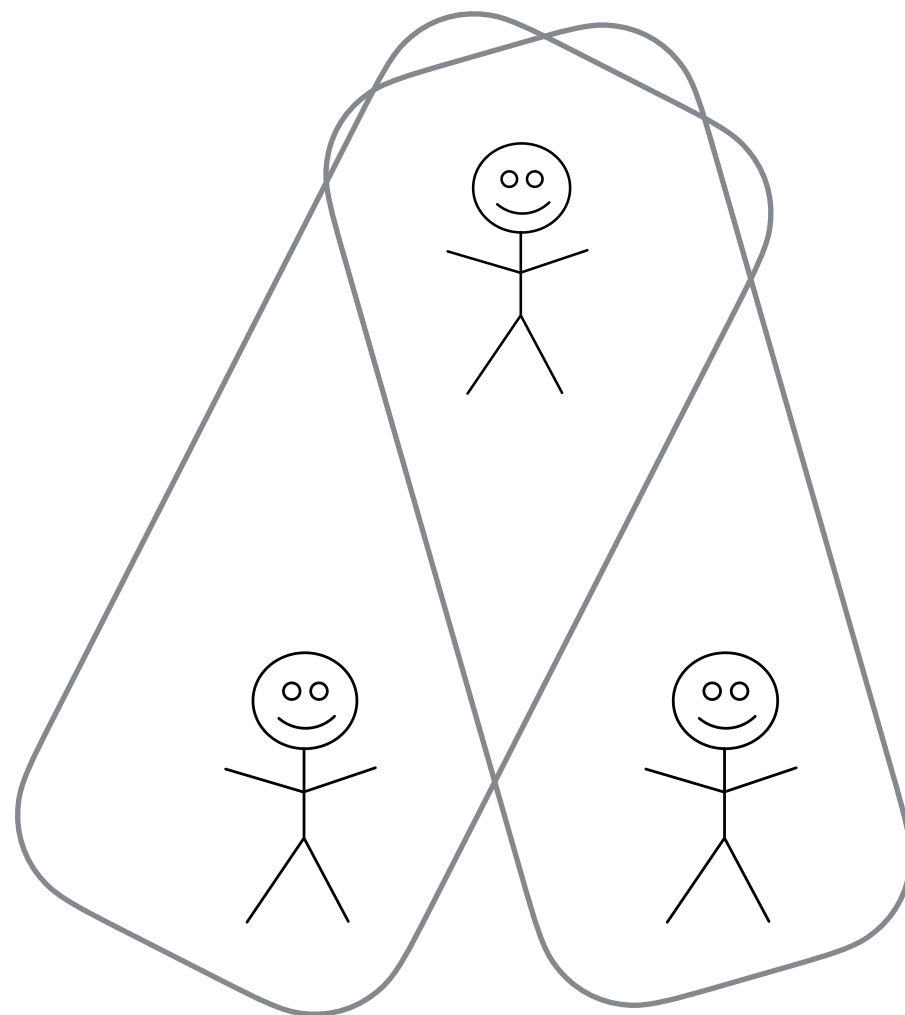
One can describe this situation in a way which makes it seem mysterious. I-90 and I-94 are separate highways; the stretch of highway north of 65th street is one highway; and yet it is identical to both I-90 and I-94. How can this be???

The puzzle here is only superficial. Roads are things which have different spatial parts located in different places. Two roads can occupy the same space just by sharing some of the same parts. No big deal.

The puzzle here is only superficial. Roads are things which have different spatial parts located in difference places. Two roads can occupy the same space just by sharing some of the same parts. No big deal.

On one view, people -- and other objects -- have a relation to time which is similar to the relation in which roads stand to space. Like roads, we seem to have spatial parts. But on this view, we also have **temporal parts**. Just as I exist in separate locations by having separate spatial parts in those distinct locations, so I exist at different times by having different temporal parts in those different times.

pre-division

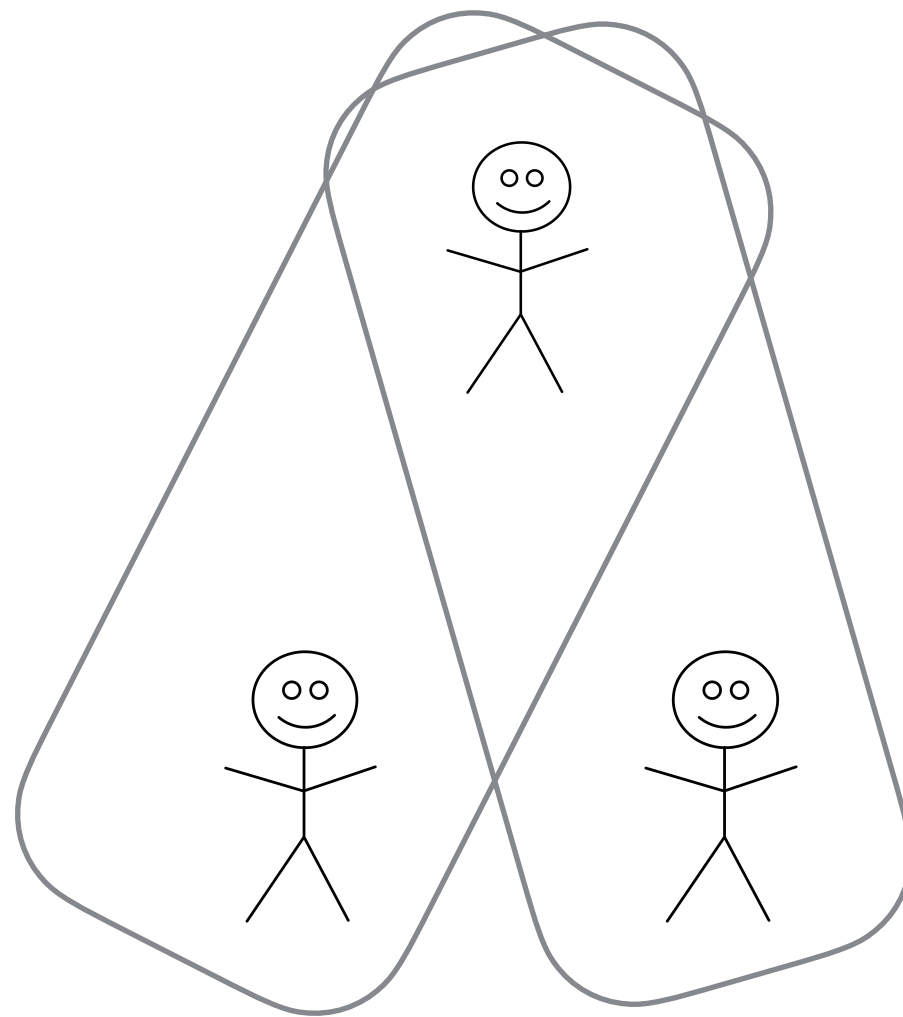


post-division

On this view, Lefty and Righty are indeed distinct. But they have a temporal part in common: you, before the surgery.

This kind of view - on which survival is a matter of having different temporal parts at different times - is called **four dimensionalism**.

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post-division

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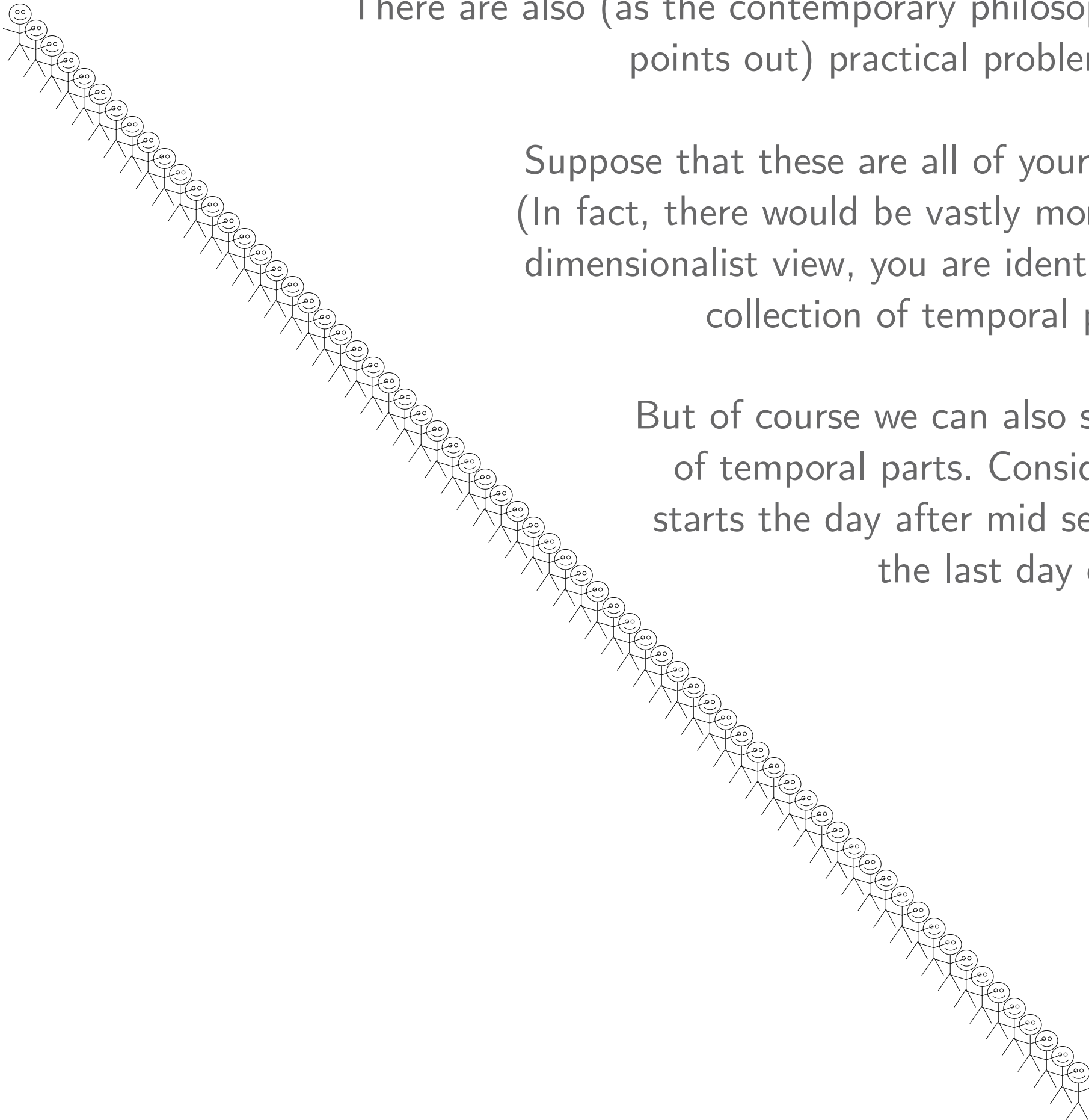
Four dimensionalism provides a way to be a materialist or psychological theorist without a non-branching clause. But it comes with a cost. You never know how many people are 'located' inside of one body. Indeed, you can't know, without knowing your future, how many people you are. This strikes some people as absurd.

This kind of view might also promise to help with the split brain cases. Perhaps individuals before the corpus callosum is severed are like the individual at the start of My Division.

There are also (as the contemporary philosopher Mark Johnston points out) practical problems.

Suppose that these are all of your temporal parts. (In fact, there would be vastly more.) On the four-dimensionalist view, you are identical to this large collection of temporal parts.

But of course we can also select smaller collections of temporal parts. Consider the collection that starts the day after mid semester break and ends the last day of exams.



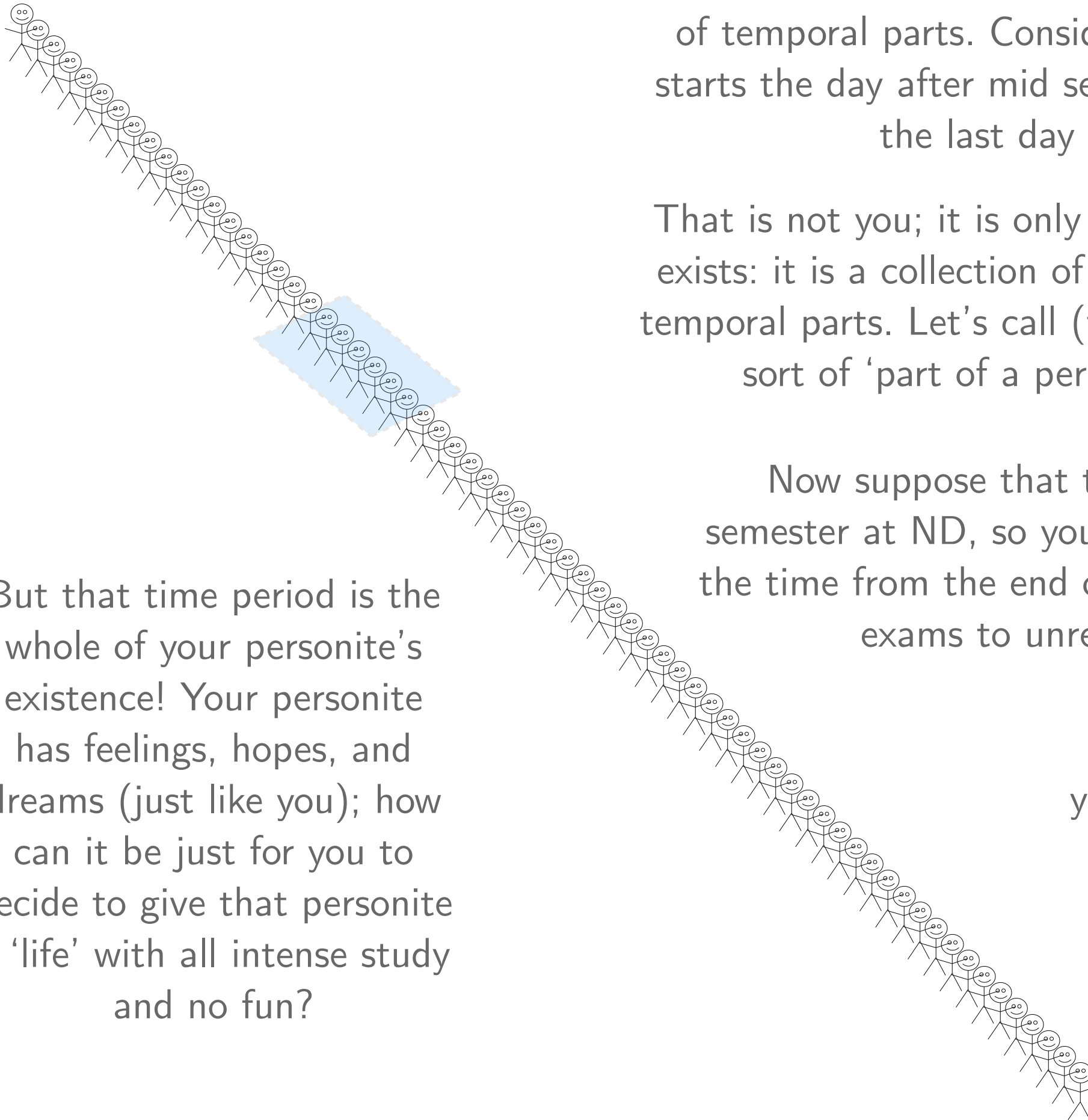
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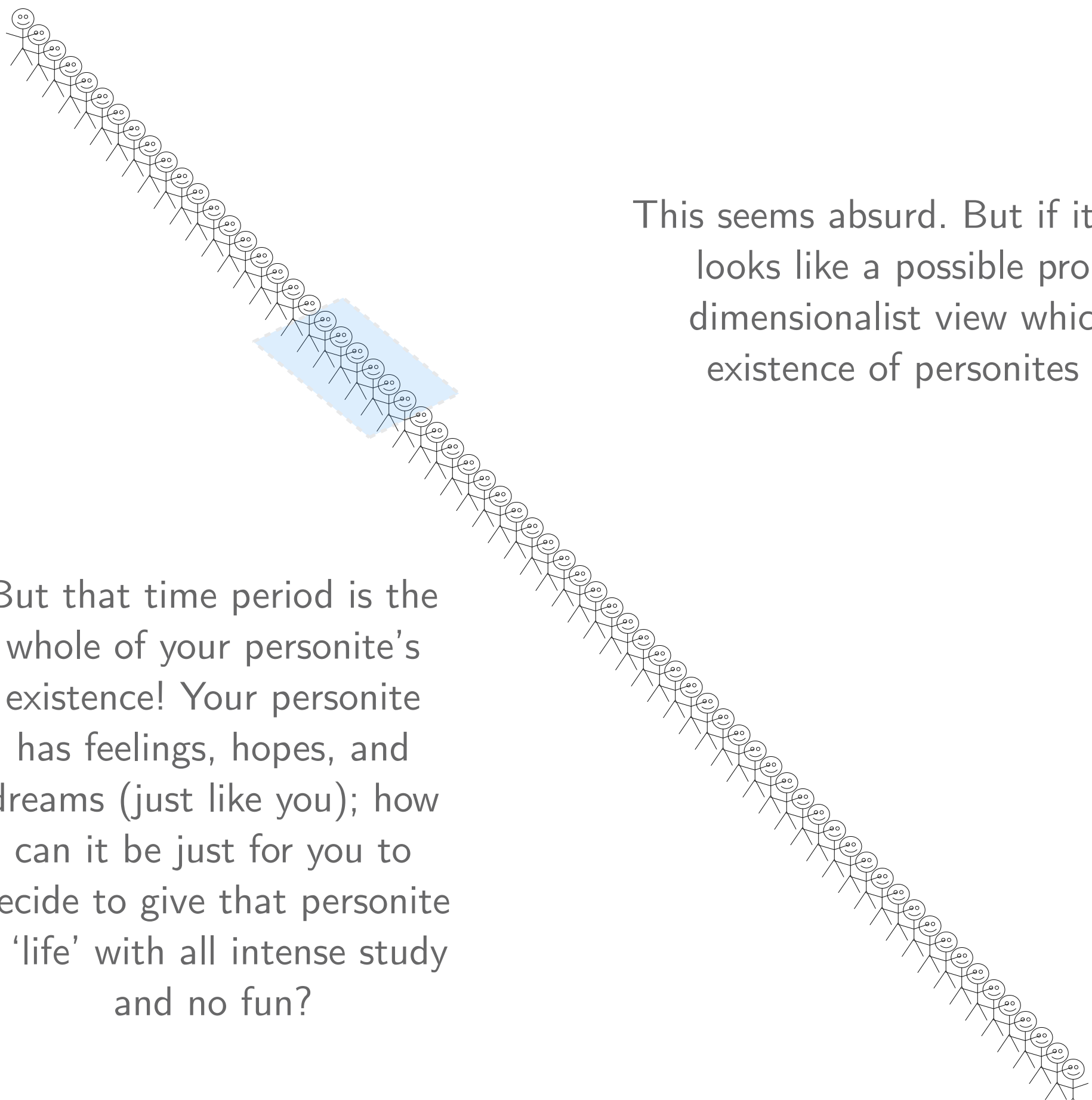
That is not you; it is only part of you. But it still exists: it is a collection of psychologically related temporal parts. Let's call (following Johnston) this sort of 'part of a person' a 'personite.'

Now suppose that this is your toughest semester at ND, so you decide to devote all of the time from the end of break until the end of exams to unremitting study.

It will be painful to push yourself so hard; but it will be worth it in end, you think.

But that time period is the whole of your personite's existence! Your personite has feelings, hopes, and dreams (just like you); how can it be just for you to decide to give that personite a 'life' with all intense study and no fun?





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This seems absurd. But if it really is absurd, that looks like a possible problem for the four-dimensionalist view which gives rise to the existence of personites in the first place.

What should the believer in souls say about My Division?

At first glance, the believer in souls might seem to have a much easier time than other views. After all, according to the believer in souls, survival is a matter of the survival of an immaterial soul. And immaterial souls (unlike brains) cannot be split into two.

But My Division can still be used to pose a challenge for the believer in souls. It seems that the believer in souls has two options:

In My Division,
your soul gets
connected to
neither of the
resulting bodies,
so that neither
Lefty nor Righty is
you.

In My Division,
your soul gets
connected to one of
Lefty or Righty
(but not both).

But there are worries about both.

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On the view that neither Lefty nor Righty is you, we face the same challenge that the materialist faced: the challenge of explaining why you survive Hemisphere Transplant but not My Division.

And the options for responding to this challenge are the same. We can deny that you survive Brain Transplant or Hemisphere Transplant — but that seems implausible.

Or we can say that neither Lefty nor Righty is you because the physical connection is branching. But that is open to the challenges faced by all “non-branching” theories.

In My Division,
your soul gets
connected to one
of Lefty or
Righty (but not
both).

Could we say instead that in this case one of Lefty or Righty would be you? Perhaps in this kind of case the psychophysical laws randomly connect the soul to one of the two hemispheres.

But suppose that you are Lefty, and you are wondering whether you are the same person as the person whose body was fatally injured. You know that Righty is wondering the same thing. At most one of you is the person you seem to remember being. Will you ever be able to discover who really is that person?

But suppose that you are Lefty, and you are wondering whether you are the same person as the person whose body was fatally injured. You know that Righty is wondering the same thing. At most one of you is the person you seem to remember being. Will you ever be able to discover who really is that person?

It seems that you will never be able to figure out which one of you really is that person. And yet this fact would seem to be a fact of great importance. Suppose that the person whose body was destroyed was married; wouldn't his spouse want to know which person they are married to?

In this scenario, there is some temptation to believe that there cannot be a very important unknown fact about who Lefty and Righty are. It seems that once we know all of the facts about the physical and psychological relations between Lefty and Right and the person whose body was destroyed, we know all of the important facts. But that would seem to leave no role for immaterial souls.